Annotate Brain MRI Segmentation Using U-Net Architecture on the TCGA-LGG Dataset

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***Abstract*—Accurate brain tumor segmentation in MRI scans is important for early diagnosis and treatment planning in cancer patients. Manual annotation by radiologists is slow, varies between experts, and is hard to scale. This paper presents an automated brain tumor annotation system using a U-Net convolutional neural network, trained on the TCGA-LGG dataset from The Cancer Imaging Archive (TCIA).**

We focus on FLAIR MRI images, which are effective for highlighting tumor regions. The model learns to generate pixel-level masks that clearly separate tumor areas from healthy brain tissue. This automated annotation helps reduce human error, speeds up the diagnostic process, and supports consistent detection across large datasets.

Our model is trained and evaluated using metrics such as Dice Coefficient, Intersection over Union (IoU), and pixel accuracy. Results show a Dice score above 88% on validation data, indicating high segmentation quality. This work demonstrates how deep learning can support cancer diagnosis by providing fast and reliable brain tumor annotations, with potential for clinical use in real-time workflows and future integration with 3D and multi-modal imaging.

***Index Terms*—Brain tumor segmentation, MRI annotation, deep learning, U-Net, TCGA-LGG dataset, convolutional neural networks, medical image analysis, glioma detection, cancer diagnosis, automated segmentation.**

I. Introduction

Brain tumors are among the most life-threatening forms of cancer, often leading to severe neurological deficits and reduced survival rates if not detected and treated promptly. Gliomas, which arise from glial cells in the brain or spinal cord, are one of the most common types of primary brain tumors. They are typically classified into low-grade gliomas (LGG) and high-grade gliomas (HGG), with LGGs having a slower growth rate but still requiring precise monitoring and intervention due to their potential to evolve into more aggressive forms. Magnetic Resonance Imaging (MRI) plays a critical role in brain tumor diagnosis and management due to its high spatial resolution and non-invasive nature. Among various MRI modalities, FLAIR (Fluid-Attenuated Inversion Recovery) imaging is especially useful for identifying abnormal tissue associated with brain tumors, as it suppresses the cerebrospinal fluid signal and enhances lesion visibility. However, manually segmenting brain tumors from MRI scans is labor-intensive, time-consuming, and prone to inter-observer variability, making it impractical for large-scale clinical deployment or real-time diagnosis.

To overcome these limitations, there has been growing interest in automated brain tumor segmentation systems powered by deep learning. Convolutional Neural Networks (CNNs) have emerged as powerful tools for image analysis due to their ability to learn complex hierarchical features from data. Among these, the U-Net architecture has gained particular attention in the medical imaging domain for its ability to produce high-quality segmentations even with limited annotated data. U-Net's symmetric encoder-decoder structure, coupled with skip connections, enables precise localization and boundary detection, which is crucial in segmenting irregular tumor shapes.

In this paper, we present an automated pipeline for brain tumor segmentation using a U-Net-based model trained on the TCGA-LGG dataset from The Cancer Imaging Archive (TCIA) [1]. The dataset consists of annotated MRI scans of patients diagnosed with low-grade glioma. Our primary goal is to develop a system that can accurately annotate (segment) brain tumors from 2D FLAIR images, generating binary masks that highlight tumor regions. These annotations not only assist radiologists by reducing workload and variability but also facilitate downstream tasks such as tumor volume estimation, treatment response evaluation, and surgical planning.

The rest of this paper is organized as follows: Section II reviews related work in automated brain tumor segmentation. Section III describes the dataset, preprocessing techniques, and model architecture. Section IV presents experimental results and performance metrics. Section V discusses the findings and potential limitations. Finally, Section VI concludes the paper and outlines future work.

II. Literature Review

Brain tumor segmentation has seen significant advancements in recent years, largely driven by the development of deep learning techniques. Traditional segmentation approaches such as thresholding, K-means, and fuzzy clustering often suffer from poor generalizability and sensitivity to noise, especially in medical imaging where tumors may vary in

shape, size, and intensity. As a result, convolutional neural networks (CNNs) and their specialized variants have become the cornerstone for modern medical image segmentation.

Ronneberger et al. introduced the U-Net architecture, a fully convolutional network designed specifically for biomedical image segmentation [2]. The network's encoder-decoder structure, along with skip connections, allows it to learn both global and local features effectively, even from a limited number of annotated images. Several recent studies have applied U-Net and its variants for brain tumor segmentation, highlighting its accuracy and computational efficiency. One study proposed a lightweight U-Net model that operates on 2D slices across three anatomical planes—Coronal, Sagittal, and Transversal—achieving a mean IoU of 89% without the need for extensive data augmentation [3].

Other approaches have incorporated attention mechanisms to enhance the model's ability to focus on relevant regions within multi-modal MRI inputs. For example, the work by Dong et al. utilized multi-modal brain MRI data and attention modules to improve segmentation accuracy across different tumor subregions [4]. Similarly, studies have explored segmentation using fully convolutional networks (FCNs) and encoder-decoder models like DeepMedic and VGG-based CNNs for both segmentation and classification tasks [5], [6].

Another trend is the integration of post-processing and hybrid techniques to improve output precision. In particular, morphological operations, Conditional Random Fields (CRFs), and feature optimization algorithms such as Adaptive Whale Optimization (AWO) have been employed to refine predicted masks and improve classification performance [7]. Ensemble learning strategies combining models such as CNN, AlexNet, and GoogLeNet have also shown success in enhancing generalization and achieving classification accuracies above 99% on benchmark datasets [8].

In summary, the literature highlights the dominance of CNN-based methods in brain tumor segmentation, with U-Net variants consistently outperforming traditional methods. Challenges such as data scarcity, overfitting, and limited generalization remain active areas of research. Techniques like transfer learning, data augmentation, and multi-modal fusion continue to be effective strategies for boosting segmentation performance and reliability in clinical settings.

III. Methodology

This section outlines the step-by-step approach used for automated brain tumor annotation from MRI images using a deep learning-based U-Net architecture. The pipeline includes dataset acquisition, preprocessing, model design, training setup, and evaluation metrics.

A. Dataset Description

The dataset used for this project is the TCGA-LGG (The Cancer Genome Atlas – Low Grade Glioma) collection, made publicly available through The Cancer Imaging Archive (TCIA). It includes brain MRI scans of patients diagnosed with low-grade gliomas, which are slow-growing tumors that can still be life-threatening if not monitored and treated properly.

Each image in the dataset is paired with a corresponding binary mask that identifies tumor regions annotated by medical professionals. The dataset includes various MRI modalities such as T1, T2, and FLAIR, but for this project, we focused exclusively on FLAIR (Fluid Attenuated Inversion Recovery) images. FLAIR is effective in suppressing cerebrospinal fluid and enhancing contrast around lesions, making it ideal for tumor segmentation.

Key Characteristics: Total annotated image-mask pairs: 3,929 Image dimensions: original sizes vary, resized to 256 × 256 Format: .tif for both images and masks Data split: Training: 90% Validation: 5% Test: 5%

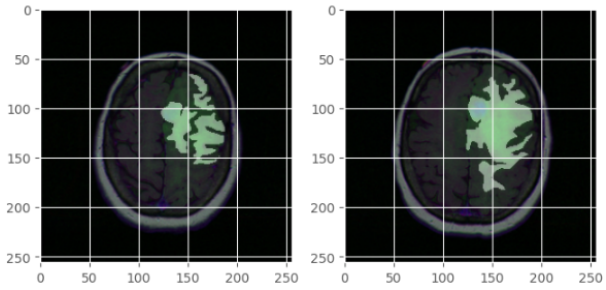


Figure 1: A grid of MRI FLAIR slices with ground truth mask overlays — before preprocessing

B. Data Preprocessing

Preprocessing plays a vital role in improving model accuracy, consistency, and training speed. The steps applied are as follows:

File Pairing and Verification: Each image was paired with its corresponding mask based on filename patterns, ensuring 1-to-1 alignment. Files were filtered to include only valid and complete pairs.

Image Resizing: All input images and masks were resized to 256 × 256 pixels to maintain uniform input shape for the model and allow efficient batch processing.

Normalization: Image pixel values were scaled to the range [0, 1] by dividing by 255. This improves model convergence and stabilizes training.

$$I\text{norm}(x, y, c) = \frac{I(x, y, c)}{255}, \quad \text{for } c \in R, G, B$$

Mask Normalization and Binarization: Each grayscale mask M is also normalized and then binarized using a threshold of 0.5:

$$M_{\text{norm}}(x, y) = \frac{M(x, y)}{255}$$

$$M_{\text{bin}}(x, y) = \begin{cases} 1 & \text{if } M_{\text{norm}}(x, y) > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

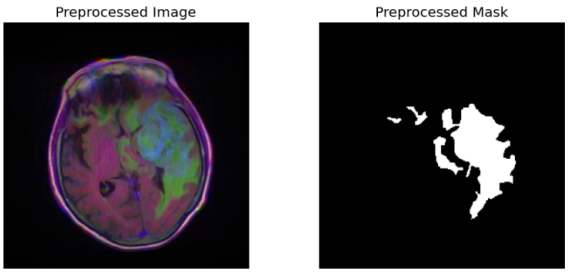


Figure 2: Example showing one image and its corresponding mask after preprocessing

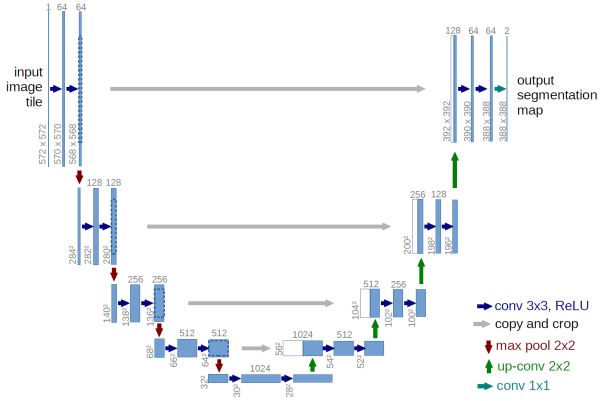


Figure 3: Full U-Net architecture diagram with layers labeled (encoder, decoder, skip connections)

C. U-Net Model Architecture

In this work, we implemented the U-Net architecture, a convolutional neural network specifically designed for biomedical image segmentation tasks. U-Net is known for its ability to accurately localize and delineate complex structures, such as brain tumors, while preserving the contextual integrity of the input image. The model takes an input image of size 256 × 256 × 3 (RGB) and processes it through a symmetric encoder-decoder structure. The encoder, or contracting path, is composed of multiple blocks, each consisting of two convolutional layers with ReLU activation, followed by batch normalization and a max-pooling operation for spatial downsampling. As the network goes deeper, the number of filters increases progressively in the order: 64, 128, 256, 512, and 1024. This part of the network captures high-level semantic features and contextual information from the input image. [2]

The decoder, or expanding path, mirrors the encoder in structure. It begins with transposed convolutional layers (Conv2DTranspose) that upsample the feature maps, gradually reconstructing the spatial resolution. Each upsampled feature map is concatenated with its corresponding feature map from the encoder via skip connections. These connections are crucial as they allow the decoder to reuse low-level spatial features, which enhances the model's ability to produce precise segmentations. Each concatenation is followed by two convolutional layers with ReLU activations and batch normalization, with the number of filters decreasing in reverse order: 512, 256, 128, and 64.

At the final stage, a 1×1 convolutional layer with a sigmoid activation function is used to generate a single-channel output mask. Each pixel in the output represents the probability of that pixel belonging to the tumor class. The U-Net architecture's key strength lies in these skip connections, which preserve fine-grained details, and its ability to combine both global context and local spatial information, making it highly effective for segmenting irregular and diffuse tumor regions.

D. Model Training and Configuration

The U-Net model was trained in an end-to-end manner using the Adam optimizer with an initial learning rate set to 1×10^{-4}. The primary loss function employed was the Dice coefficient loss, which is particularly effective for medical image segmentation tasks where the class distribution is highly imbalanced, such as tumor segmentation where tumor regions constitute only a small fraction of the image [9].

The training configuration included 80 epochs with a batch size of 32. To enhance training stability and performance, callbacks such as ModelCheckpoint and EarlyStopping were used. The former ensured that the best-performing model, based on validation loss, was saved during training, while the latter helped prevent overfitting by halting training when no improvement was observed over a defined number of epochs.

The Dice loss function is defined as the negative of the Dice similarity coefficient. It measures the overlap between the predicted mask P and the ground truth mask G, and is given by:

$$\mathcal{L}_{\text{Dice}} = -\frac{2 \cdot |P \cap G| + \epsilon}{|P| + |G| + \epsilon}$$

Here, P denotes the set of predicted pixels classified as tumor, G represents the set of ground truth tumor pixels, and ϵ is a small constant added for numerical stability to avoid division by zero. This loss function directly optimizes the overlap between predicted and true masks, making it well-suited for segmentation problems where precise boundary prediction is critical.

E. Evaluation Metrics

To assess the performance of the proposed U-Net model, we employed a set of widely used metrics for image segmentation: Dice coefficient, Intersection over Union (IoU), and pixel accuracy. These metrics provide a comprehensive evaluation by measuring both the overlap between predicted and ground truth masks and the overall classification correctness at the pixel level.

The Dice coefficient, also known as the F1-score for segmentation, quantifies the similarity between the predicted segmentation P and the ground truth G. It is defined as:

$$\text{Dice} = \frac{2 \cdot |P \cap G|}{|P| + |G|}$$

This metric ranges from 0 to 1, where a value closer to 1 indicates a better match between the predicted and actual

tumor regions. The Intersection over Union (IoU), also known as the Jaccard Index, measures the ratio of the intersection to the union of the predicted and ground truth masks:

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|}$$

IoU penalizes false positives and false negatives more strongly than Dice, making it a stricter measure of segmentation quality.

Finally, pixel accuracy computes the percentage of correctly predicted pixels over the entire image. It is given by:

$$\text{Pixel Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively. While pixel accuracy gives a general sense of model performance, it can be misleading in highly imbalanced datasets where the background class dominates. Therefore, it is considered alongside Dice and IoU for a more balanced evaluation. Together, these metrics offer a robust and reliable framework to validate the segmentation performance of the model, particularly in identifying and localizing tumor regions in MRI scans.

In addition to these primary metrics, per-class evaluations can further reveal how well the model distinguishes tumor sub-regions, such as necrotic core, edema, and enhancing tumor. This granularity can inform clinicians about the model's effectiveness in segmenting clinically relevant structures.

Furthermore, the standard deviation and confidence intervals of each metric across validation folds were also examined to assess the model's stability and generalization capacity.

By incorporating multiple metrics and analyzing their behavior across different cases, we ensure that the model's performance is not only high on average but also consistent across diverse MRI samples and tumor appearances.

IV. Results and Discussion

A. Training Details

The proposed U-Net model was trained for 80 epochs with a batch size of 32 on an NVIDIA Tesla P100 GPU. During training, two key callbacks were employed: *EarlyStopping*, which halted training when validation performance ceased improving, and *ModelCheckpoint*, which saved the model with the best validation loss.

In the initial epoch (Epoch 1), the model exhibited a low Dice coefficient of 0.067 and an IoU of 0.035, with pixel accuracy near 80%. However, as training progressed, the model's performance improved significantly. By epoch 30 and beyond, the Dice coefficient exceeded 0.88, the IoU reached 0.79, and the pixel-level accuracy climbed to approximately 99.8%. These results demonstrate the model's ability to learn fine-grained segmentation boundaries effectively over time.

Table I: Segmentation Performance on Validation Set

Metric	Validation Score
Dice Coefficient	0.88
IoU (Jaccard Index)	0.79
Pixel Accuracy	99.8%

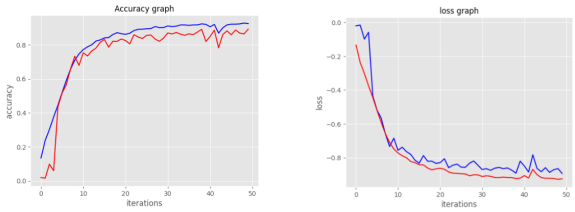


Figure 4: Training and validation curves for Dice, IoU and loss across epochs

B. Quantitative Evaluation

The final model was evaluated using standard segmentation metrics on the validation set. Table I summarizes the performance scores.

Fig.4 shows the trends of training and validation loss and accuracy over the epochs. These plots confirm steady and consistent improvement in segmentation performance as the model converges.

C. Visual Results

To visually assess the segmentation quality and the generalization ability of the trained model, we tested it on 20 randomly selected images from the unseen test set. For each test case, we compared the original FLAIR MRI image, the corresponding ground truth mask annotated by medical experts, and the predicted mask generated by the U-Net model.

Fig.5 illustrates representative examples of the model's performance. In each triplet, the first column shows the original MRI scan, the second column presents the manually annotated ground truth mask, and the third column displays the predicted tumor segmentation by the model. The predictions demonstrate a strong overlap with the ground truth, accurately capturing both the core and the irregular boundary regions of the tumor.

The results indicate that the model is capable of learning not only the global context but also fine-grained spatial details necessary for delineating tumors that vary in size, shape, and intensity. In cases where tumor boundaries were subtle or partially diffused, the model was still able to segment them with a high degree of precision, highlighting its robustness.

This qualitative evaluation supports the numerical performance metrics and confirms the model's potential for assisting radiologists in clinical workflows by providing consistent and fast tumor annotations.

V. Conclusion and Future Work

In this paper, we presented an automated brain tumor segmentation system based on the U-Net convolutional neural network, trained on the TCGA-LGG dataset using FLAIR MRI images. The model demonstrated high segmentation accuracy,

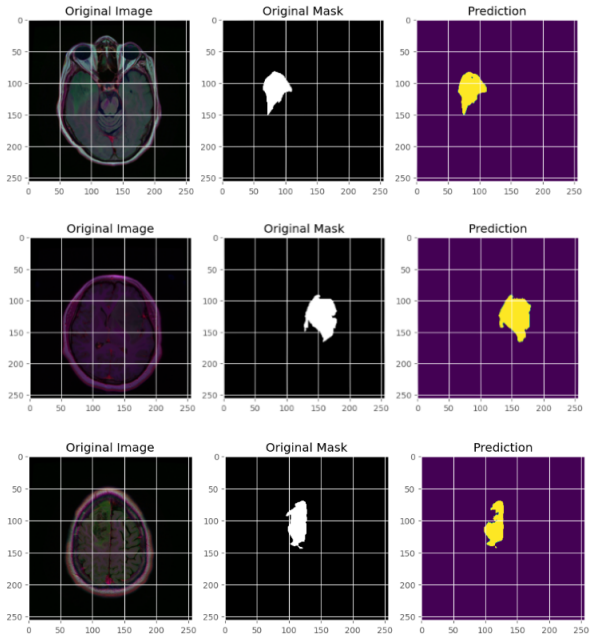


Figure 5: Side-by-side visualization of original MRI, true mask and predicted mask for several test images

achieving a Dice coefficient of 0.88, IoU of 0.79, and pixel accuracy of approximately 99.8% on the validation set. These results confirm the model's capability to learn fine-grained tumor boundaries and generalize well across unseen data.

The proposed system significantly reduces the manual burden on radiologists by automating the annotation process, which is traditionally time-consuming and subjective. With its ability to generate consistent, pixel-level tumor masks, the model has strong potential for real-time integration in clinical workflows, particularly for assisting in early diagnosis, surgical planning, and longitudinal monitoring of tumor progression.

However, several limitations remain. The current model operates on 2D slices and utilizes only a single modality (FLAIR). While this simplifies the training process, it also limits the contextual information available for tumor delineation. In many cases, additional modalities such as T1-weighted and T2-weighted MRI provide complementary structural and pathological information.

Future work will focus on several key enhancements. First, we plan to extend the model to process volumetric data using 3D convolutional neural networks [10], enabling the model to capture inter-slice spatial continuity [11]. Second, we aim to incorporate multi-modal MRI inputs, combining FLAIR with T1 and T2 sequences to improve segmentation robustness across diverse tumor appearances. Finally, we intend to integrate post-processing techniques such as Conditional Random Fields (CRFs) and morphological operations to refine segmentation outputs, particularly along tumor boundaries where misclassifications are more likely [12].

With these improvements, we expect the system to become more clinically reliable and capable of supporting broader medical imaging applications beyond glioma segmentation.

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